

CSE220
Online on Convolution

Subsections:

Total Marks: 10

Total Time: 30 Minutes

Two cascaded LTI systems have the equivalent impulse response obtained by convolving their individual impulse responses. Consider

$$h_1[n] = \delta[n] - \delta[n - 1] = [1, -1] \quad (\text{first difference}), \quad h_2[n] = u[n] \quad (\text{running-sum accumulator}).$$

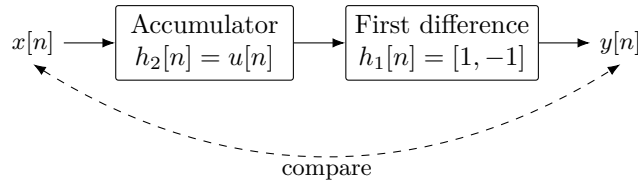
The accumulator has zero initial condition and computes

$$v[n] = \sum_{k=-\infty}^n x[k].$$

Since

$$(h_1 * h_2)[n] = u[n] - u[n - 1] = \delta[n],$$

the cascade should act as the identity system. Instead of explicitly constructing $h_1 * h_2$, you will verify this using the non-impulse input signal given below and compare the final output with the original input.



In this online, reuse your `DiscreteSignal` and `LTISystem` classes from the offline. Represent both systems by their impulse responses and connect them using repeated calls to `LTISystem.output`.

Given

$$\begin{aligned} x[n] : \quad n &= -2 \dots 8, & \text{values} &= [2, -1, 3, 1, -2, 0, 0, 0, 0, 0], \\ h_1[n] : \quad n &= 0 \dots 1, & \text{values} &= [1, -1], \\ h_2[n] : \quad h_2[n] &= 1 \text{ for } n \geq 0. \end{aligned}$$

Finite-window note. The infinite sequence $u[n]$ cannot be stored. In the template, store $h_2[n] = 1$ for $n = 0 \dots 10$. This is enough for every sample of the given input to affect the complete graded observation window $n = -2 \dots 8$. Compare signals only on that window.

Tasks

- Implement `cascade(first_system, second_system, input_signal)`. Apply `first_system.output` first, then `second_system.output`, and return both the intermediate and final outputs.
- Implement `max_absolute_difference_in_range(first, second, start, end)` over the inclusive range `start...end`.
- Construct $x[n]$, $h_1[n]$, and the stored window of $h_2[n]$ using `DiscreteSignal`; then create the two `LTISystem` objects.
- Apply $x[n]$ through **Accumulator** $\rightarrow$ **First difference**. Print the input samples, accumulator-response samples, and first-difference-response samples for $n = -2 \dots 8$.
- Using the plotting helper provided in the template, plot $x[n]$, the accumulator response $v[n]$, and the first-difference response $y[n]$ over $n = -2 \dots 8$ as discrete-time stem plots.
- Compute and print the maximum absolute difference between the first-difference response and $x[n]$ over $n = -2 \dots 8$.
- In a print statement, conclude that the cascade output equals $x[n]$, so the cascade is an identity system and the differentiator and accumulator are inverse systems under zero initial conditions.

Restriction

Do not use `numpy.convolve`, `scipy.signal`, or any built-in convolution routine.

Submission

Rename your Python file as 2305xxx.py and submit it in Moodle.